

Learning Targets

Upon successful completion of this course, you will be able to say the following:

Algebra

A1 Solving equations and inequalities I can use algebraic operations to solve equations and evaluate inequalities.

A2 Factoring, multiplying, and rational expressions I can rewrite rational and polynomial expressions through factoring, polynomial multiplication, and simplifying fractional expressions.

Functions

F1 Inputs and outputs I can use function notation, find and simplify the output of a function given an input, and find an input that produces a given output; I can do this for functions given by a formula, a table, and a graph. (1.1 & 1.2)

F2 Rate of change I can find the average rate of change in a function on a given interval and interpret the meaning. (1.3)

F3 Linear and quadratic functions I can model linear and quadratic relationships with equations and find key features graphically and algebraically. (1.4 & 1.5)

F4 Compositions I can find the composition of functions given graphs, tables, and contexts, and interpret the meaning (1.6)

F5 Inverses I can determine whether a function has an inverse and find the formula of the inverse of basic functions that have an inverse. (1.7)

F6 Translations and piecewise functions I can translate graphs functions and construct piecewise functions. (1.8 & 1.9)

F7 Function review I can interpret, represent, and evaluate problems using all previous function learning objectives.

Trigonometry

T1 Unit circle and arc length I can generate graphs of a circular functions given appropriate information about the circle being traversed, and describe key features of the function. (2.1, 2.2, & 2.3)

T2 Sinusoidal functions I can find the amplitude, period, and midline of a transformed version of the basic sine or cosine function; and I can find the formulas for transformed versions of the basic sine or cosine function that have certain described properties. (2.4)

T3 Right Triangles Given a right triangle with partial information about its sides and angles, I can determine the missing information for the remaining sides and angles, even if the quantities involve an unknown variable. (4.1)

T4 Finding Angles I can interpret inverse trigonometric functions and use them to find angles. (4.3 & 4.4)

T5. Other Trig Functions I can define tangent and reciprocal trigonometric functions and their graphs. (4.2 & 4.5)

T6 Trig Review I can interpret, represent, and evaluate problems using all previous exponential and log learning objectives.

Exponentials and Logarithms

E1 Exponent Properties I can simplify exponential expressions and solve equations using properties of exponents. (3.0)

E2 Exponential Functions I can construct graphs and equations of exponential and logarithmic functions given contextual data or two point values. (3.2 & 3.2)

E3 Equations with Exponents and Logs I can simplify logarithmic expressions using properties of logarithms and find exact solutions to equations involving exponential and logarithmic expressions. (3.4, 3.5, 3.6)

E4 Exponential Review I can interpret, represent, and evaluate problems using all previous exponential and log learning objectives.

Rational Functions and Polynomials

R1 Polynomials and Power Functions I can identify the degree, zeros, turning points, and long-range behavior of a polynomial given its formula or graph, and I can find the formula of a polynomial that fits given data about its behavior. (5.1, 5.2, & 5.3)

R2 Rational Functions I can identify important features of rational functions, such as the asymptotes, end behavior, intercepts and holes. I can find the formula of a rational function that fits given data about its behavior. (5.4 & 5.5)

Participation and Engagement Goals

- Coming to class prepared by watching preview videos, taking the video quizzes on time, and being ready to learn while in class. Above 90% is highly proficient, above 75% is proficient.
- Coming to class on time, working problems, and sharing ideas in class. Participating in out of class enrichment. Above 90% is highly proficient, above 75% is proficient.
- Completing WebWork assignments on time. Above 90% is highly proficient, above 75% is proficient.

- Exceptional persistence in the face of challenges.